



Ottobot 2.0 delivering orders to a traveller at an airport (Courtesy: Ottobot.io)

claim that their technologies will be feature-ready this year. Implementing autonomous long-haul deliveries is every truck maker's dream. As driving on highways is a monotonous task the drivers would be happy to be relieved of it, unlike short, local trips.

In a demonstration by Kodiak Robotics and U.S. Express, last March, a self-driving eighteen-wheeler travelled non-stop for five days, carrying goods between Dallas and Atlanta. It did four round trips, which means more than 10,000km, in five days. It would have taken ten days for a human driver to do, with regular breaks in between. The trip was, of course, supervised by drivers, taking turns behind the wheel. And according to reports by the company, they had to intervene more than once.

Although highways are more structured and easier to manoeuvre than city roads, there is still a lot for trucks to learn, including changing lanes at the right time and knowing when to stop. Hence, even the so-called fully-autonomous trucks will continue to have drivers in them in the near future.

Another class of autonomous trucks is for middle- and last-mile delivery, that is, for transporting goods between warehouses or to local stores. This requires travelling on

public roads, which is quite a challenge.

Microsoft is planning to invest more than \$10 million in Gatik, a California based company that is actively championing the use of autonomous trucks on public roads.

Its customers

include Walmart, Georgia-Pacific, KBX, and Pitney-Bowes. They use medium-duty, Level-4 ready trucks from Isuzu. Gautam Narang, Co-Founder and CEO of Gatik, said in a recent interview that they expect to have hundreds of trucks deployed in commercial operations by the end of 2023.

Udelv is another California based company in this space. Its Transporter is a cabless autonomous delivery vehicle with a modular cargo pod that can carry a total of 900kg of goods. The company targets middle- and last-mile delivery. Travelling at speeds up to 112kmph, Udelv's vehicle can cover between 250 and 500km depending on the battery pack, and make up to 80 stops per cycle. Udelv hopes to have 50,000 vehicles operating on public roads by 2028.

In December 2022, Carrefour Drive and Goggo Network collaboratively trialled mobile delivery using an autonomous delivery vehicle operating on public roads, both in heavy traffic and in normal conditions, covering 15km at up to 70kmph. Online orders are packed and loaded on the shuttle, which makes its way to customers who can open their locker using a password shared through SMS to retrieve the goods. The shuttle is capable of making multiple stops in one run.

Carrefour has obtained authorisation to run its vehicle on a circuit of 15 kilometres in France.

Since they are heavy and can prove hazardous in case of a malfunction, the self-driving trucks—on highways as well as public roads—are still highly regulated and are permitted only in a few regions across the world. Forbes contributor and trend watcher, Richard Bishop, writes, “2024 appears to be the launch window for commercial driverless on highways, if statements by Aurora and Kodiak play out according to plan. Therefore, 2023 will see a substantial portion of driverless operations in conducting final on-road validation testing. OEMs will keep their launch dates close to the chest, but I'd be very surprised if they made a move sooner. Waymo marches to their own drummer, and if I had to place a bet, I'd put my money on them to be the first mover here, maybe even during the coming year.”

Ground delivery bots

This is where ground delivery bots and small vehicles have an advantage. Regulatory bodies that are wary of large trucks are more open to the idea of small autonomous delivery vehicles and bots. These are four- or six-wheeled vehicles or robots that travel on sidewalks or streets to deliver food and parcels to end consumers. Sensors enable them to assess obstacles and safely navigate around people, animals, and objects. They are small, move slowly, and carry small payloads.

Restaurants, grocery stores, and other establishments partner with service providers to use their app based robots for deliveries within a small radius. When a customer places an order, a robot from the network is assigned to pick up their order from their vendor and deliver it to a pinned location. In most cases, the delivery bot has a locker